

WASECA, MN, USA

WMO: 722032

Lat: 44.074N

Lon: 93.553W

Elev: 343

StdP: 97.27

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.0	-27.8	0.3	-23.8	-25.2	0.4	-21.8	14.3	-5.8	12.8	-7.6	2.4	240	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.1	22.6	29.1	21.6	27.7	20.6	24.4	28.6	23.4	27.5	22.5	26.4	5.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.3	27.2	22.3	17.7	26.6	21.2	16.6	25.4	75.5	29.2	71.7	28.0	68.1	26.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.7	9.2	-28.5	34.7	3.0	1.8	-30.7	36.0	-32.4	37.1	-34.1	38.1	-36.4	39.4		
			-28.7	26.7	3.0	1.6	-30.9	27.8	-32.6	28.7	-34.3	29.6	-36.5	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-8.7	-7.6	0.3	7.9	14.3	19.6	21.3	19.8	16.8	8.9	1.7	-5.9
	DBStd	12.02	7.20	6.74	6.80	6.05	5.11	3.37	3.10	3.00	4.47	5.18	6.00	6.36
	HDD10.0	2338	579	493	307	109	17	0	0	0	3	83	254	493
	HDD18.3	4261	837	726	558	316	147	24	8	18	80	296	499	752
	CDD10.0	1408	0	0	8	44	149	289	352	304	207	49	6	0
	CDD18.3	290	0	0	1	1	21	63	102	64	34	4	0	0
	CDH23.3	2616	0	0	2	28	236	553	936	514	302	45	1	0
	CDH26.7	614	0	0	0	3	75	134	236	91	67	8	0	0
Wind	WSAvg	4.2	4.8	4.6	4.8	5.2	4.8	3.9	3.0	2.6	3.4	4.0	4.6	4.5
Precipitation	PrecAvg	884	26	24	56	78	105	131	115	122	86	66	48	32
	PrecMax	1122	63	56	102	157	193	216	183	259	289	165	143	91
	PrecMin	583	5	3	8	21	41	80	44	44	25	16	5	6
	PrecStd	137	15	13	23	36	43	42	46	47	60	42	34	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	11.2	22.2	27.0	32.8	32.8	32.9	31.2	31.1	27.6	20.2	11.0
		MCWB	3.0	8.9	15.4	15.1	19.4	22.5	25.6	23.7	21.5	17.2	13.5	8.2
	2%	DB	3.6	5.2	17.1	22.8	27.9	30.0	31.1	28.8	28.0	23.0	16.0	5.3
		MCWB	2.0	3.3	12.1	14.3	17.2	20.7	24.0	23.4	20.4	16.8	10.3	3.3
	5%	DB	2.2	2.6	12.8	20.3	26.0	27.7	28.9	27.4	27.0	21.1	12.6	3.0
		MCWB	1.1	1.6	9.8	12.6	16.5	18.9	23.1	21.8	19.7	14.8	8.8	1.9
	10%	DB	1.0	1.3	9.9	17.3	22.7	26.3	27.6	26.2	25.0	17.4	10.9	2.2
		MCWB	0.2	0.4	7.6	10.9	15.0	18.5	21.8	21.0	18.9	12.0	7.4	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	8.8	16.8	16.9	21.6	24.2	27.0	25.2	23.6	21.5	14.1	7.6
		MCDB	5.4	11.1	21.6	22.6	28.6	29.9	32.0	29.2	28.2	25.0	17.7	10.0
	2%	WB	2.1	3.6	12.9	15.3	19.6	22.7	25.1	23.9	22.4	17.1	11.6	3.7
		MCDB	3.4	5.1	15.6	21.7	25.9	27.1	29.5	27.5	26.6	22.2	14.9	5.3
	5%	WB	1.0	1.6	9.8	13.6	18.0	21.6	23.9	23.0	21.4	14.9	9.0	2.3
		MCDB	1.8	2.6	13.2	18.8	23.7	26.0	28.0	26.6	25.1	19.2	12.8	3.2
	10%	WB	0.0	0.4	6.8	11.8	16.4	20.3	22.8	22.0	20.0	12.7	7.1	1.1
		MCDB	0.6	1.0	9.5	16.1	21.5	24.7	26.8	25.4	23.9	17.2	9.9	1.9
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	8.5	10.7	11.3	10.8	11.0	10.9	11.7	11.0	9.0	7.7
		MCDBR	8.5	9.1	13.1	15.6	15.0	12.8	11.5	11.5	13.4	15.4	13.6	8.3
		MCWBR	7.6	7.7	9.0	9.1	7.1	6.2	6.5	6.8	7.0	8.8	8.8	7.2
	5% WB	MCDBR	8.1	8.7	11.8	13.5	11.9	10.7	10.5	10.0	11.1	12.8	12.6	7.9
		MCWBR	7.5	7.5	8.5	8.7	6.6	6.7	6.6	6.6	6.8	8.4	8.6	6.9
Clear-Sky Solar Irradiance	taub		0.267	0.287	0.313	0.368	0.394	0.410	0.412	0.404	0.374	0.323	0.289	0.267
	taud		2.435	2.370	2.422	2.314	2.304	2.300	2.292	2.305	2.359	2.499	2.513	2.454
	Ebn at Noon		874	909	928	894	878	862	855	849	848	858	840	838
	Edh at Noon		72	92	100	122	127	128	128	122	107	81	67	64
All-Sky Solar Radiation	RadAvg		1.77	2.85	3.72	4.60	5.39	6.02	6.37	5.34	4.23	2.75	1.78	1.30
	RadStd		0.21	0.30	0.37	0.49	0.43	0.48	0.27	0.38	0.30	0.36	0.16	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page